

8. Cystic fibrosis is an example of an inherited disease caused by a recessive allele. A carrier of the disease has a genotype Ff.

- (a) Compare the chances of a child being born with cystic fibrosis if both parents are carriers, with a child whose father suffers from cystic fibrosis and a mother who is a carrier. [4]

- (b) Cystic fibrosis causes a build-up of thick mucus in the lungs, which leads to poor gas exchange. Mucus in the lungs also allows increased bacterial growth that is responsible for infections.

Explain how the body defends itself against bacterial infection in the lungs. [3]

- (c) Sufferers of cystic fibrosis benefit from being physically active to slow the rate of decline in lung function.

Explain the short-term and long-term effects of exercise on lung function. [3]

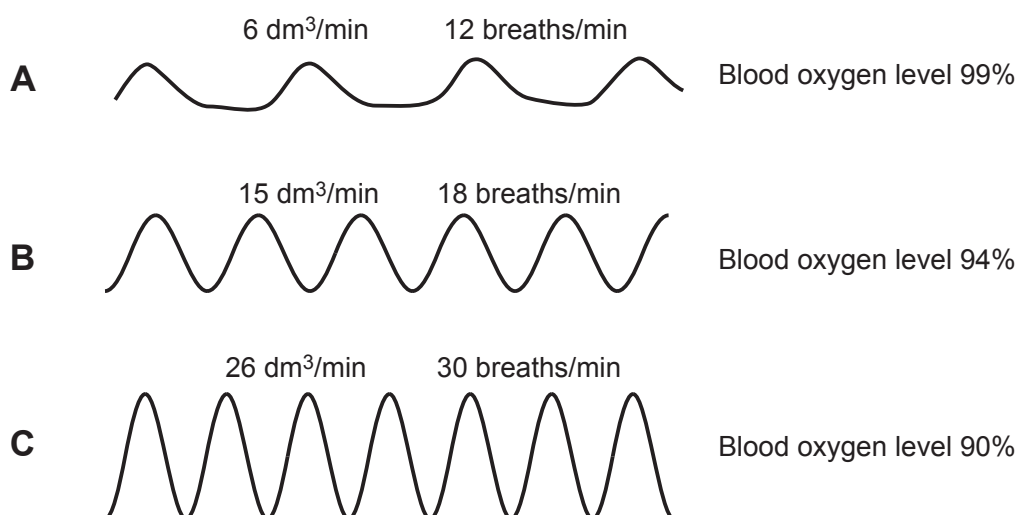


- (d) The diagram below gives information about the breathing of three different individuals, all of whom are resting.

A is a person who is not affected by cystic fibrosis.

B is a person with cystic fibrosis.

C is a person with cystic fibrosis and a lung infection.



The numbers show the volume of air breathed in per minute in dm³ per min and also the number of breaths per minute.

- (i) Use the equation:

$$\text{volume of air per breath} = \frac{\text{volume of air breathed in per minute}}{\text{number of breaths per minute}}$$

to compare the volume of air breathed in per breath for each of the breathing patterns.

[4]

A B C

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- (ii) Compare the number of breaths per minute with blood oxygen levels in the three individuals. [2]

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END OF PAPER

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THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1																	
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	4 He Helium 2
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89	Key														

1 H Hydrogen 1

Key

A_r	relative atomic mass
Symbol Name	
Z	atomic number